

Enterprise Autonomous Job-Hunting Architecture & Technical Integration Specification

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3. Zero-Authentication Public APIs Selection

Per strict operational constraints, all integrated external APIs must be **100% free, requiring zero registration, zero API keys, and zero auth tokens**. Below are the verified public APIs from the official public-apis/public-apis repository chosen for CareerHero Studio:

Category	API Name & Endpoint	Auth	CareerHero Studio Integration Role
Job Data	Arbeithnow documenter.getpostman.com/...	None	Direct REST feed for verified remote & European tech/design postings. Ingests full descriptions without anti-bot blocks.
Job Data	freehire freehire.dev/docs/api	None	Open-source aggregator querying direct company ATS boards (Greenhouse, Lever, Workable). Bypasses job board middlemen.
Job Data	RemoteOK API remoteok.com/api	None	Zero-auth live JSON feed of worldwide remote product design, software, and AI roles.
Job Data	Remotive API remotive.com/api/remote-jobs	None	Categorized remote jobs feed with direct application links, salary brackets, and location tags.
Email Audit	Kickbox Open API open.kickbox.com/v1/verify	None	Validates hiring manager emails, verifies MX records, and flags disposable inboxes before sending outreach to prevent domain burn.
Email Audit	Disify API disify.com/api/email/{email}	None	Zero-auth disposable email detector ensuring recruiter leads are authentic corporate addresses.
Company Intel	Pick an Agency pickanagency.com/developers	None	Directory of 47,000+ creative, product design, and digital agencies by location and rating. Feeds CareerHero's Agency queue.
Company Intel	Domainsdb.info api.domainsdb.info	None	Domain search verifying corporate web domains and canonical DNS records for newly discovered startup leads.
Geography	REST Countries restcountries.com	None	Country classification (UAE, UK, US, India) used to enforce regional CV rules (photo vs no-photo, phone conventions).
Geography	Nominatim (OSM) nominatim.org/api	None	Global forward/reverse geocoding normalizing city variants (e.g. Bangalore vs Bengaluru, Dubai vs UAE).
Currency	Frankfurter API frankfurter.app/docs	None	ECB exchange rates normalizing multi-currency compensation (GBP, USD, EUR, INR, AED) to uniform target CTC.

4. Awesome-LLM-Apps Integration Roadmap

Shubham Saboo's awesome-llm-apps repository provides battle-tested agent patterns that solve critical friction points in CareerHero Studio:

- **Project Graveyard (Installed locally):** Adapted into an *Application Autopsy Engine*. Automatically classifies stalled outreach (>14 days unreplied) by failure pattern: 'ATS Keyword Wall', 'Ghosted After First Touch', or 'Over-Seniority Mismatch'.
- **First Reader (Agent Skill):** Pre-dispatch simulation testing. Analyzes cold email drafts and portfolio case studies using synthetic hiring manager personas to highlight cognitive drop-off points, jargon bloat, and weak hooks.
- **Advisor-Orchestrator-Worker Loop:** Heavy-duty reasoning split across models. Claude/Gemini-Pro acts as the *Advisor* formulating the strategic angle; Gemini 3.8 Flash acts as the high-throughput *Worker* generating 20 tailored variations in seconds.
- **Always-On Background Daemon Pattern:** Modeled after the *Always-on HN Briefing Agent*, running a lightweight continuous cron task that polls fresh job boards every 6 hours, filters for target roles, and prepares a daily staging queue.

5. D:\JobSpy (AuraJobs) Engine Integration

The standalone engine at D:\JobSpy contains advanced multi-board scraping, rate-limit handling, query expansion, and role classification. CareerHero Studio connects directly to this engine via a native service bridge rather than calling detached batch scripts.

AuraJobs Core Components & Adapters

The engine operates 10 distinct source adapters orchestrated by a balanced scheduler:

- **MultiBoardAdapter:** Aggregates LinkedIn, Indeed, Glassdoor, Google, Bayt, and Naukri with regional balancing (India, Middle East, Global).
- **ATSAdapter:** Direct scraping of career boards built on Greenhouse, Lever, and AshbyHQ, yielding high-conversion non-indexed listings.
- **Direct API Adapters:** RemoteOK, Remotive, Himalayas, Arbeitnow, FreeHire, and AIJobs delivering instant JSON structured records.
- **Pipeline Processors:** RoleClassifier (enforces title matching and negative term exclusion), VisaDetector (relocation/sponsorship parsing), Deduper (SHA256 fingerprinting on normalized title + company), and MatchScorer.

Continuous Ingestion Bridge Implementation (Python)

CareerHero Studio bridges D:\JobSpy by wrapping its core scheduler inside a non-blocking FastAPI background task with real-time SQLite writes:

```
# job-agent/src/services/jobspy_service.py
import sys from pathlib import Path JOBSPY_DIR = Path(r"D:\JobSpy")
if str(JOBSPY_DIR) not in sys.path: sys.path.insert(0, str(JOBSPY_DIR))
from core.scheduler import BalancedScheduler
from core.classifier import RoleClassifier
from core.normalizer import normalize_dataframe
from core.deduper import deduplicate_jobs
from core.scorer import MatchScorer
from src.web.db import ingest_scraped_batch
async def run_autonomous_scrape(role: str, geo: str = "All", freshness_hours: int = 72):
    scheduler = BalancedScheduler(role=role, geo=geo, freshness_hours=freshness_hours)
    raw_df = await scheduler.execute_async()
    cleaned_df = normalize_dataframe(raw_df)
    deduped_df = deduplicate_jobs(cleaned_df)
    # Commit directly to CareerHero SQLite
    leads_queue.ingest_count = ingest_scraped_batch(deduped_df.to_dict(orient="records"))
    return {"status": "SUCCESS", "new_leads_ingested": ingest_count}
```

Automated Ingestion Lifecycle & State Machine

```
[Cron / Webhook / UI Button] ■ ▼ [Search Profile Built] ■■> (Target: Role, Seniority, Geography, Negative Exclusions)
■ ▼ [Concurrent Adapters] ■■> (ATS Boards, Zero-Auth APIs, JobSpy Aggregators)
■ ▼ [Data Normalizer] ■■> (Standardize Title, Currency, Age Bucket, Location)
■ ▼ [Fuzzy Deduplicator] ■■> (Company + Title Fingerprint Hash Check against SQLite)
■ ▼ [Scoring & Visa Parser] ■■> (CTC Match, Taxonomy Relevance, Visa Sponsorship Flag)
■ ▼ [Atomic SQLite Insert] ■■> (Batch write to 'leads' and 'ingestion_runs' tables)
```

6. Relational Database Schema & Architecture

The application relies on an enterprise-grade SQLite schema housed at `job-agent/tracker/jobhunt.db`. The schema isolates candidate preferences, raw ingestion streams, curated leads, application dispatches, and recruiter interactions.

Table Name	Primary Keys & Constraints	Key Attributes & Types	Relational Purpose & Indexing
leads	id INTEGER PRIMARY KEY UNIQUE(company, role)	company (TEXT), role (TEXT), category (TEXT), contact_email (TEXT), website (TEXT), match_score (REAL), visa_status (TEXT), status (TEXT)	Master target queue (590+ leads). Filtered by Category (Direct, YC, Agency, Scraped). Indexed on status, category.
applications	id INTEGER PRIMARY KEY FOREIGN KEY -> leads	company (TEXT), role (TEXT), resume_path (TEXT), email_body (TEXT), gmail_message_id (TEXT), applied_date (TEXT), follow_up_date (TEXT)	Audit trail of all live and staged dispatches. Powers the 7-day follow-up detector. Indexed on applied_date.
inbound_replies	id INTEGER PRIMARY KEY gmail_message_id UNIQUE	sender_name, sender_email, snippet, intent (TEXT), confidence (REAL), suggested_status (TEXT), matched_company	Classified recruiter responses from Gmail scanner (Interview, Assessment, Rejection, Follow-up).
ingestion_runs	id INTEGER PRIMARY KEY	run_timestamp, target_role, geo, total_scraped, new_leads_added, duplicates_skipped, runtime_seconds	Telemetry and audit logging for AuraJobs scraping runs and zero-auth API polling.
profile	id PRIMARY KEY (id=1)	full_name, email, portfolio_links (JSON), resume_markdown, extracted_skills (JSON), activation_passkey, dry_run (INT)	Candidate single source of truth. Guarded by local encryption and strictly excluded from Git.

7. Detailed Database Sequence Flows

A. Automated Scheduled Refresh Sequence

[illegible]

B. Autonomous 1-Click Application & Outreach Dispatch Flow

[illegible]

8. Phased Implementation Roadmap

To reliably execute this transformation without service interruption, the work is organized into four sequential engineering phases:

Phase	Milestone & Objectives	Deliverables & Verification Gates
Phase 1 Zero-Auth Integration	Connect lightweight zero-auth APIs (Arbeitnow, Remotive, Kickbox) to CareerHero Studio.	• Add src/services/public_apis.py client. • Integrate Kickbox open verification on lead email edits. • Verify zero-auth feeds populate SQLite cleanly.
Phase 2 JobSpy Service Bridge	Mount D:\JobSpy core pipeline into CareerHero FastAPI backend as an asynchronous service.	• Create jobspy_service.py wrapper. • Add POST /api/scrapper/trigger endpoint in FastAPI. • Test non-interactive runs emitting to leads table.
Phase 3 Autonomous Scheduler	Deploy background auto-refresh daemon with cron/heartbeat support.	• Implement background worker using Python asyncio or APScheduler. • Support configurable interval (4h, 8h, 24h). • Zero-duplicate guarantee via SHA256 lead fingerprints.
Phase 4 Agentic Skills Integration	Embed first-reader attention auditor and multi-agent drafting loops from Awesome-LLM-Apps.	• Add pre-send email scoring widget in UI. • Hook Project Graveyard autopsy reports into stalled outreach tab. • Automated end-to-end Playwright UI audit.

9. Verification & Safety Protocols

CareerHero Studio strictly enforces the core guidelines outlined in AGENTS.md:

- **Zero Privacy Leak:** Workstation SQLite database (jobhunt.db), OAuth tokens (token.json), and credentials (credentials.json) remain local-first and strictly gitignored.
- **Safe Mode by Default:** DRY_RUN=True is mandatory for all new scraping pipelines and outreach batches. Live Gmail dispatches require explicit user toggle and passkey confirmation.
- **Anti-Bot Integrity:** Scraping adapters respect robots.txt and HTTP rate limits. No CAPTCHA bypassing is attempted; failed sources fail gracefully without crashing the pipeline.
- **Visual & Layout Verification:** Every generated CV is strictly budgeted to a single page using ReportLab's Bahnschrift/Helvetica flowable calculations, verified before dispatch.